

MACHINE LEARNING FUNDAMENTALS SERIES

Elastic Net Regression: Fundamentals and Properties

Combining L1 (Lasso) and L2 (Ridge) Penalties for High-Dimensional & Correlated Data

Source: StatQuest with Josh Starmer — Regularization Part 3: Elastic Net Regression

1. Introduction and Motivation

In high-dimensional machine learning workflows (such as genomics, gene expression profiling, or finance), datasets frequently contain hundreds or thousands of predictor variables ($p \gg n$). In such scenarios, practitioners face a fundamental dilemma:

- **Uncertainty in Feature Quality:** It is usually unknown in advance whether the dataset contains mostly useful predictors (where Ridge Regression excels) or a few key signals buried in massive noise (where Lasso Regression excels).
- **Limitations of Single Regularization Models:**
 - *Pure Lasso* (L_1) performs feature selection by setting coefficients to zero, but it struggles with correlated predictors (selecting one arbitrarily) and cannot select more than n features when $p > n$.
 - *Pure Ridge* (L_2) shrinks coefficients smoothly and handles correlated features well, but retains all p variables, creating complex, non-sparse models.

Elastic Net Regression solves this by combining both L_1 (Lasso) and L_2 (Ridge) penalties into a unified hybrid loss function, leveraging the strengths of both approaches simultaneously.

2. Mathematical Formulations

2.1. General Dual-Penalty Formulation

Elastic Net minimizes the Sum of Squared Residuals (SSR) subject to both absolute (L_1) and squared (L_2) coefficient penalties:

$$\mathcal{L}_{\text{ElasticNet}}(\beta) = \underbrace{\sum_{i=1}^n \left(y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j x_{ij} \right) \right)^2}_{\text{Sum of Squared Residuals (SSR)}} + \underbrace{\lambda_1 \sum_{j=1}^p |\beta_j|}_{\text{Lasso } L_1 \text{ Penalty}} + \underbrace{\lambda_2 \sum_{j=1}^p \beta_j^2}_{\text{Ridge } L_2 \text{ Penalty}} \quad (1)$$

where $\lambda_1 \geq 0$ controls L_1 sparsity and $\lambda_2 \geq 0$ controls L_2 coefficient shrinkage.

Important Rule on Intercept β_0 : Penalty terms $\sum |\beta_j|$ and $\sum \beta_j^2$ start at $j = 1$, explicitly **excluding the y-intercept** β_0 . Penalizing β_0 would artificially force baseline predictions towards zero regardless of the natural center of y .

2.2. Re-parameterized (λ, α) Mixing Formulation

Standard machine learning packages (e.g., `scikit-learn`) parameterize Elastic Net using an overall penalty scale λ and an L_1 -ratio mixing hyperparameter $\alpha \in [0, 1]$:

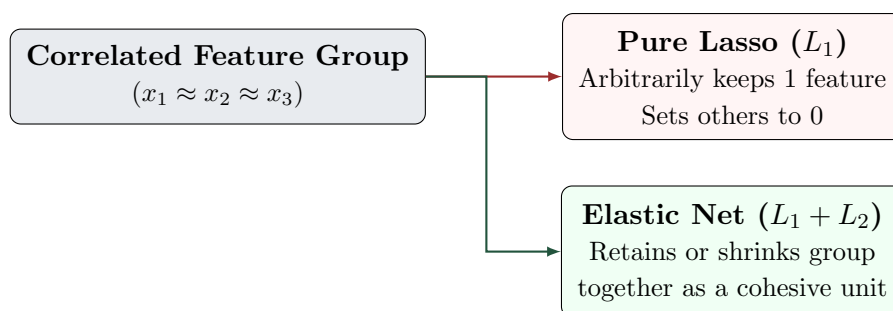
$$\mathcal{L}_{\text{ElasticNet}}(\beta) = \text{SSR} + \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + (1 - \alpha) \sum_{j=1}^p \beta_j^2 \right] \quad (2)$$

where $\alpha = 1$ yields pure Lasso, $\alpha = 0$ yields pure Ridge, and $0 < \alpha < 1$ produces a true Elastic Net hybrid.

3. Key Properties and Unique Behaviors

3.1. Grouping Effect for Correlated Features

When predictors are strongly correlated ($x_1 \approx x_2 \approx x_3$), Lasso tends to pick a single variable at random and set the rest to zero ($\beta_2 = \beta_3 = 0$). Elastic Net's L_2 component enforces a **Grouping Effect**: correlated features receive similar coefficient weights, and the entire group is either retained or eliminated together.



3.2. Overcoming $p > n$ High-Dimensional Limits

In ultra-high-dimensional settings where feature count exceeds sample size ($p > n$):

- Pure Lasso can select at most n variables before saturating.
- Elastic Net's strictly convex L_2 term allows it to select more than n variables while maintaining stable optimization.

4. Hyperparameter Tuning via 2D Grid Search & 10-Fold CV

Elastic Net relies on two hyperparameters: λ_1, λ_2 (or λ, α). Because optimal values cannot be known *a priori*, they are evaluated systematically via **10-Fold Cross-Validation**:

1. Define a 2D candidate grid of (λ, α) pairs.
2. Split training data into 10 equal folds.
3. For each (λ, α) pair, fit the Elastic Net model on 9 folds and compute validation error (SSR) on the held-out fold; repeat across all 10 folds.
4. Select the optimal pair (λ^*, α^*) that achieves the lowest mean validation error.

5. Summary Comparison of Regularization Characteristics

The table below provides a comprehensive comparison of model characteristics across Ordinary Least Squares, Ridge, Lasso, and Elastic Net Regression.

Characteristic	OLS Regression	Ridge (L_2)	Lasso (L_1)	Elastic Net ($L_1 + L_2$)
Penalty Term	None	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $	$\lambda_1 \sum \beta_j + \lambda_2 \sum \beta_j^2$
Coefficient Shrinkage	None	Shrinks towards 0 (never exact 0)	Shrinks to exact 0	Shrinks smoothly & sets exact 0
Feature Selection	No (Keeps all p)	No (Keeps all p)	Yes (Automatic sparse model)	Yes (Automatic sparse model)
Correlated Features	Unstable / High Variance	Shrinks weights smoothly	Selects 1 arbitrarily, drops rest	Groups correlated features together
High-Dim ($p > n$)	Fails / Non-invertible	Solves cleanly ($\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}$)	Selects max n features	Selects $> n$ grouped features
Hyperparameters	None	λ (1D CV)	λ (1D CV)	λ_1, λ_2 or λ, α (2D CV)
Best Use Case	Small p , no noise	Many useful predictors	Few useful predictors + noise	Unknown mix / Correlated features